

Cervical cancer

3 July 2026 | Reading time: 6 min (1578 words)

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Key facts

- Cervical cancer is largely preventable through HPV vaccination and regular screening, as recommended by national guidelines, and it can be cured if detected early and treated promptly.
- Cervical cancer is the fifth most commonly diagnosed cancer in women globally with around 604 000 new cases and around 280 000 deaths in 2024.
- The highest rates of cervical cancer incidence and mortality are in low- and middle-income countries. This reflects major inequities driven by lack of access to national HPV vaccination, cervical screening and treatment services and social and economic determinants.
- Cervical cancer is caused by persistent infection with human papillomavirus (HPV). Women living with HIV are 6 times more likely to develop cervical cancer compared to women without HIV.
- Countries worldwide are accelerating efforts to eliminate cervical cancer, guided by the global 90–70–90 targets: 90% of girls fully vaccinated with HPV vaccine by age 15, 70% of women screened by ages 35 and 45, and 90% of women with pre-cancer or invasive cancer receiving appropriate treatment.

Overview

Globally, cervical cancer is the fifth most commonly diagnosed cancer in women, with 604 000 new cases estimated in 2024. Cervical cancer is the leading cause of cancer death in women in 26 countries, primarily in sub-Saharan Africa and parts of South and Central America, Melanesia, and South-East Asia (1). These regional differences reflect inequalities in access to vaccination, screening and treatment services.

They are further influenced by risk factors such as HIV prevalence and by broader social and economic determinants, including gender inequality and poverty.

Women living with HIV are six times more likely to develop cervical cancer compared to the general population, and an estimated 5% of all cervical cancer cases are attributable to HIV (2). Cervical cancer disproportionately affects younger women, and as a result, 20% of children who lose their mother to cancer do so due to cervical cancer (3).

Causes

Almost all cases of cervical cancer are caused by infection with oncogenic types of human papillomavirus (HPV). Human papillomavirus (HPV) is a common sexually transmitted infection which can affect the skin, genital area, anal area and throat. Almost all sexually active people will be infected at some point, usually without symptoms. In most cases, the immune system clears the virus naturally. Persistent infection with certain carcinogenic types of HPV can cause abnormal cells that may develop into cancer.

Persistent HPV infection of the cervix (the lower part of the uterus or womb, which opens into the vagina – also called the birth canal) can lead to precancerous lesions which if left untreated cause about 95% of cervical cancers. It usually takes 15–20 years for abnormal cells to become cancer. In women with weakened immune systems, such as untreated HIV, this process can be faster and take 5–10 years. Factors that increase the risk of cancer progression include: the grade of oncogenicity of the HPV type, immune status, the presence of other sexually transmitted infections, number of births, young age at first pregnancy, hormonal contraceptive use, and smoking.

Prevention

Boosting public awareness, strengthening health literacy, and improving access to information and services are key to prevention and control across the life course:

- **HPV vaccination for girls 9–14 years is highly effective at preventing infection, cervical cancer and other HPV-related cancers.**
- **Cervical screening from the age of 30 (25 years in women living with HIV) can detect cervical precancer, and when coupled with timely treatment can prevent progression to cervical cancer.**
- **At any age, early detection of women with symptoms, followed by prompt quality treatment, can cure cervical cancer.**

HPV vaccination and other prevention steps

As of 2025, there are 8 licensed HPV vaccines, five of which have received WHO pre-qualification and are available globally. All these protect against the high-risk HPV types 16 and 18, which cause around 76% of cervical cancers.

HPV vaccination is a priority for all girls aged 9–14 years, before they become sexually active. Depending on the national schedule the vaccine may be given as one or two doses. Individuals with compromised immune systems, including people living with HIV, should ideally receive two or three doses. Some countries have additionally chosen to vaccinate boys to further reduce the prevalence of HPV in the community and to prevent cancers in men caused by HPV.

Other important ways to prevent HPV infection and reduce the risk of cervical cancer include:

- being a non-smoker or stopping smoking
- using condoms
- voluntary male circumcision.

Cervical screening and treatment of precancers

Women should be screened for cervical cancer with a high-performance test every 5–10 years starting at age 30. [Women living with HIV should be screened every 3–5 years, starting at age 25.](#) The global strategy encourages a minimum of two lifetime screens with a high-performance test by age 35 and again by age 45. Precancers rarely cause symptoms, which is why regular cervical cancer screening is important, even if you have been vaccinated against HPV.

Self-collection of a sample for HPV testing, which may be a preferred choice for women, has been shown to be as reliable as samples collected by healthcare providers.

After a positive screen, a health-care provider can look for changes on the cervix (such as precancers) which may develop into cervical cancer if left untreated. Treatment of precancers is a simple and effective procedure to prevent cervical cancer. Treatment may be offered in the same visit for screening (the screen-and-treat approach) or after a second test (the screen, triage and treat approach), which is especially recommended for women living with HIV.

Treatment of precancerous lesions is usually quick and may involve limited discomfort compared to other medical procedures. The process involves examining the cervix after applying acetic acid, with or without magnification (colposcopy or naked-eye visual

inspection), to locate the lesion and determine the appropriate treatment. Treatment options include:

- thermal ablation which uses a heated probe to destroy abnormal cells;
- cryotherapy which uses a cold probe to destroy abnormal cells;
- LEEP or LEETZ (large loop excision of the transformation zone) which uses an electric heated loop to remove abnormal tissue; and/or
- cone biopsy which uses a surgical instrument to remove a cone-shaped piece of tissue for further assessment.

Early detection, diagnosis and treatment of cervical cancer

Cervical cancer can be cured if diagnosed and treated at an early stage of disease. Recognizing symptoms and seeking medical advice to address any concerns is a critical step. Women should consult a health-care professional if they notice:

- unusual bleeding between periods, after menopause, or after sexual intercourse
- increased or foul-smelling vaginal discharge
- symptoms like persistent pain in the back, legs, or pelvis
- weight loss, fatigue and loss of appetite
- vaginal discomfort
- swelling in the legs.

Clinical evaluations and diagnostic tests are essential for confirming cervical cancer. These are generally followed by referral for treatment services, which can include surgery, radiotherapy and chemotherapy, as well as palliative care to provide supportive care and pain management.

Management pathways for invasive cancer care are important tools to ensure that a patient is referred promptly and supported as they navigate the steps to diagnosis and treatment decisions. Features of quality care include:

- a multidisciplinary team ensuring diagnosis and staging (histological testing, pathology, imaging) takes place prior to treatment decisions;
- treatment decisions in line with national guidelines; and
- interventions are supported by holistic psychological, spiritual, physical and palliative care.

As low- and middle-income countries scale-up cervical screening, more cases of invasive cervical cancer will be detected, especially in previously unscreened populations. Therefore, referral and cancer management strategies need to be implemented and expanded alongside prevention services.

WHO Response

All countries have made a commitment to eliminate cervical cancer as a public health problem. The WHO [Global strategy](#) defines elimination as reducing the number of new cases annually to 4 or fewer cases per 100 000 women and sets three targets to be achieved by the year 2030 to put all countries on the pathway to elimination in the coming decades:

- 90% of girls vaccinated with the HPV vaccine by age 15;
- 70% of women screened with a high-performance test by 35 years of age and again by 45 years of age; and
- 90% of women with cervical precancer or cancer receiving treatment.

[Modelling estimates](#) that achieving the elimination goal could avert 74 million new cases of cervical cancer and prevent 62 million deaths by 2120, [with additional analyses highlighting the impact among women living with HIV](#).

Prevention of HPV-associated precancer and cancer is also a key element of WHO's [Global health sector strategy on HIV, hepatitis and sexually transmitted infections 2022–2030](#), and the World Health Assembly resolution WHA74.5 (2021) on oral health includes actions on mouth and throat cancers.

World Cervical Cancer Elimination Day

17 November is observed as [World Cervical Cancer Elimination Day](#) to strengthen global efforts to prevent and treat cervical cancer. The day highlights the importance of HPV vaccination, screening and treatment, with a focus on supporting women and girls. It encourages countries, WHO and partners to collaborate, expand services and track progress, building on the Global Strategy to eliminate cervical cancer.

References

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2. Stelze, Dominik et al. Estimates of the global burden of cervical cancer associated with HIV. *The Lancet*. 2020. [https://doi.org/10.1016/S2214-109X\(20\)30459-9](https://doi.org/10.1016/S2214-109X(20)30459-9)
3. Guida, F., Kidman, R., Ferlay, J. *et al*. Global and regional estimates of orphans attributed to maternal cancer mortality in 2020. *Nat Med* 28, 2563–2572 (2022). <https://doi.org/10.1038/s41591-022-02109-2>

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